

JIANGTAO KONG

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🎓 EDUCATION

College of William & Mary, VA, Williamsburg 09/2021 – Now

MS/PhD Computer Science

Beijing University of Technology, Beijing, Beijing 09/2015 – 07/2019

Bachelor Electronic Information Engineer

📖 PUBLICATIONS

- ECA: Efficient Continual Alignment for Open-Ended Image-to-Text Generation. Under Review
- Hybrid Memory Replay: Blending Real and Distilled Data for Class Incremental Learning arXiv
- YoooP: You Only Optimize One Prototype per Class for Non-Exemplar Incremental Learning arXiv
- DSAM: A Distance Shrinking with Angular Marginalizing Loss for High Performance Vehicle Re-identification arXiv
- SPATIAL TEMPORAL ATTENTIONAL GLIMPSE FOR HUMAN ACTIVITY CLASSIFICATION IN VIDEO ICIP
- Global And Local Deep Feature Representation Fusion for Vehicle Re-Identification VCIP

👤 INTERNSHIPS/PROJECTS

JPMORGANCHASE& CO. (New York) 06/2024 – 08/2024

Part Time (Virtual) 09/2024 – 12/2024

Summer Associate in Research Technologies Program Leader: Shaohan Hu

Summer Research Associate: Research in Continual Learning for Multi-modal Models

- Participated in the development of the algorithm for models to continually learn to generate the caption for different visual input for different groups.
- Proposed an algorithm for continual learning in multi-modal models and filed for intellectual property.
- ECA: Introducing a new notion of continual alignment, which incrementally adapts the alignment module within pre-trained VLMs to preserve high-quality cross-modal representations. Based on this idea, we propose Efficient Continual Alignment (ECA), a novel non-exemplar IL approach for open-ended image-to-text generation (OpenITG) tasks. Additionally, we construct four new Incremental Learning (IL) OpenITG benchmarks that better reflect real-world scenarios. Experimental results demonstrate that ECA significantly mitigates catastrophic forgetting and improves IL performance compared to baseline methods.

**Institute of Automation, Chinese Academy of Science,
National Laboratory of Pattern Recognition (Beijing)**

10/2020 – 06/2021

Internship Leader: Jun Wan

Research Intern - Student

- Participated in the development and design of face attribute annotation system.
- Participated in the Huawei ModelZoo, CV group, converted and transplanted the existing face detection models (Retina Face, FaceBoxes, etc.) to the Huawei cloud platform.

Pensees (Beijing) (Beijing)

10/2019 – 04/2020

Internship Leader: ShengMei Shen

Algorithm researcher: Research Vehicle re-ID to improve re-ID accuracy

- Participated in Vehicle Re-identification, VCIP 2019, won second place

- **DSAM Loss:** Presenting a novel Distance Shrinking with Angular Marginalizing (DSAM) loss function to perform hybrid learning in both the Original Feature Space (OFS) and the Feature Angular Space (FAS) by using the local verification and the global identification information, respectively.

Experimental Results: The DSAM loss can be used with Softmax loss, and improves the performance in both the Vehicle re-ID tasks and Person re-ID tasks. The mAP on the PKU-VD-1-Large Dataset is increased by 10.41%, the mAP on the PKU-VehicleID Dataset is increased by 9.34%, and the mAP on the VeRi-776 Dataset is increased by 8.73%; In the Person re-ID task, the mAP increased by 5.10% on DukeMTMC-reID, 2.78% on Market1501, and 6.60% on CUHK03-Detected.

Paper Link: <https://arxiv.org/abs/2011.06228>

**Institute of Automation, Chinese Academy of Science,
Intelligent System and Engineering Research Center (Beijing)**

07/2018 – 07/2019

Internship Leader: Junliang Xing

Research Intern - Student

- Researched the basic algorithms of reinforcement learning: Q-Learning, A2C, A3C, etc.
- Participated in AIIDE StarCraft AI Competition. The main task: Test and analyze.
- Participated in the development of SCII AI. The main task: Write action macros, debug engineering bugs, and participate in discussion strategies

Project Link:

<https://github.com/Snowball0823/pysc2-rl>

<https://github.com/witwolf/pysc2-rl> -> branch:test, macro-active

Hybrid Memory Replay

06/2023 – 06/2024

Project Exemplar-based Incremental Learning

- In this work, we optimized a hybrid memory including both real and synthetic data. Specifically, we propose an innovative modification to Data Distillation that distills synthetic data from a sliding window of checkpoints in history (rather than checkpoints on multiple training trajectories). Conditioned on the synthetic data, we then optimize the selection of real exemplars to provide complementary improvement to the DD objective.
- **Experiment Results:** For CIFAR-100, the average incremental accuracy (AIA) of different baselines with our hybrid memory under the zero-base-5-phases and the zero-base-10-phases: iCaRL+H: 69.47%, 60.00%, BEEF+H: 76.66%, 71.04%, FOSTER+H: 78.79%, 76.01%. For TinyImageNet, the AIA under the zero-base-5-phases and the zero-base-10-phases: iCaRL+H: 62.97%, 50.78%, BEEF+H: 69.54%, 64.06%, FOSTER+H: 70.28%, 68.55%.

Paper Link: <https://doi.org/10.48550/arXiv.2410.15372>

YoooP

12/2022 – 06/2023

Project Non-Exemplar-based Incremental Learning

- In this work, we developed two non-exemplar-based methods, YoooP and YoooP+, for class-incremental learning. Specifically, YoooP only needs to store and replay one optimized prototype for each class without generating synthetic data from stored prototypes. As an extension of YoooP, YoooP+ proposed to create synthetic data from the stored prototypes and the stored distribution of cosine similarity with the help of a high-dimensional rotation matrix. The evaluation results on multiple benchmarks demonstrated that both YoooP and YoooP+ can significantly outperform the baselines in terms of accuracy and forgetting. Importantly, this work offered a new perspective on optimizing class prototypes for exemplar-free CIL.
- **Experiment Results:** For CIFAR-100, the average incremental accuracy (AIA) under the zero-base-5-phases: YoooP: 64.90%, YoooP+: 67.26%. The AIA under the zero-base-10-phases: YoooP: 57.66%, YoooP+: 61.93%. For TinyImageNet, the AIA under the zero-base-5-phases: YoooP: 57.23%, YoooP+: 59.99%. The AIA under the zero-base-10-phases: YoooP: 53.24%, YoooP+: 58.00%.

STAG for Human Activity Classification in Video

05/2018 – 11/2018

Personal Project Human action classification in videos

- We propose a new module to deal with long-term dependency problems in human action classification tasks. In the module, we propose two blocks with small numbers of parameters to be learned for long-term dependency capturing and attention feature capturing. Consequently, when utilizing the existing model as the backbone, our model presents both robust performance and effectiveness of computation cost. Meanwhile, several extended module designs show a great potential for better performance.
- **Experiment results:** For comparing with the existing long-term extraction model (Non-Local Network), the STAG model keeps equal accuracy on top-1 and top-5 with Non-Local Network with less computational cost. The flops of the Non-Local Network is 37.9B, $1.54 \times$ Flops of I3D; the flops of STAG is 31.2B, $1.26 \times$ Flops of I3D. On the Kinetics dataset, the STAG model outperforms I3D by around 3% on top-1 and 1.6% on top-5.

Paper Link: <https://ieeexplore.ieee.org/abstract/document/8803586>

⚙️ SKILLS

- Research Interests: Deep Learning→Continue Learning, Multi-modal Models
- Languages: Python > C > BASH > C++ > PHP, SQL, JavaScript
- OS: Ubuntu, Kali, MacOS, Windows
- DL Frameworks: PyTorch, TensorFlow

♥️ AWARDS

<i>Second team</i> , Vehicle Reidentification by VCIP	2019
<i>Meritorious Winner</i> , Mathematical Contest In Modeling	2018
<i>Third Awards</i> , Lanqiao Competition Beijing Group A (C/C++)	2018
<i>Bronze Award</i> , Asian Zone, Qingdao Station, ACM International Collegiate Programming Contest	2017
<i>First Prize</i> , Annual TIS Science and Technological Innovation and Practice	2017
<i>Second Prize</i> , 9th "Challenge Cup" National College Student Curricular Academic Science and Technology Works Competition	2017